# Does erg performance follow a 1.064 power law?

Evidence from verified Concept2 RowErg rankings, season 2026

Generated 2026-10-30

## 1 Summary

We tested whether, for the same athlete in the same season, race time grows with distance as $t \propto d^{1.064}$. The analysis uses 3,993 athlete-seasons (17,446 verified results) from the Concept2 RowErg rankings, covering both sexes and all age groups.

- **On average the exponent is slightly above 1.064.** The pooled within-athlete exponent is 1.067 [1.066, 1.068]. Using 1.064 instead of 1.067 changes a predicted 6 km time from a 2 km time by 0.35% and a predicted 10 km time by 0.51%.
- **Does it vary between athletes?** The pooled exponent is 1.085 for ages 19–29 and 1.060 for 60+, and this age gradient remains inside fixed distance windows and in the exponent at 2 km (1.088 vs 1.064). Women minus men is −0.009 over all ages.
- **It depends on distance.** The local exponent at 2 km is 1.071 [1.070, 1.072] and it falls by −0.0100 per doubling of distance (curvature −0.0072). A single power law therefore fits a limited range of distances well and the pooled slope depends on which events are included.
- **Sub-maximal efforts are a limitation.** Across five alternative treatments of unusual results the pooled exponent moves by at most 0.003 from the baseline (Section 5.5), but the direction of any bias is unknown.

## 2 Hypothesis and model

For one athlete, the hypothesis is that time depends on distance as a power law, $t = k\,d^{\beta}$ with $\beta = 1.064$. Equivalent forms are average pace $\propto d^{\beta-1} = d^{0.064}$ and average power $\propto d^{-3(\beta-1)} = d^{-0.192}$. For each athlete-season $i$ with results $j$ we fit

$$\ln t_{ij} = a_i + b \ln\left(\frac{d_{ij}}{2000\ \text{m}}\right) + \varepsilon_{ij},$$

and test $b = \beta$. Timed events (4, 30 and 60 minutes) enter as the pair (fixed time, distance rowed); distance events as (nominal distance, time). To look for departures from a straight line we add a quadratic term, $\ln t = a_i + b\,x + c\,x^2$ with $x = \ln(d\ /\ 2000)$. Then $b$ is the local exponent at 2 km and $2c$ is how fast the local exponent changes with $\ln d$.

## 3 Data

**Source.** Concept2 logbook rankings (log.concept2.com/rankings), RowErg only, season 2026 (the Concept2 season, not the calendar year). We used the "verified only" lists and kept individual results that carry an athlete identifier. Race-verified rows mostly lack an identifier and could not be linked reliably across events. Each ranking lists a person's best verified result for an event and season, so athletes appear once per event.

**Events.** The events at the extremes (1 minute, 100 m, half marathon, marathon, 100 km) were left out on purpose: they are dominated by anaerobic capacity, pacing and endurance limits that no single power law is expected to cover. The remaining events span roughly 500 m to 10 km (4 to 60 minutes), as in Table 1.

Table 1: Events used. "In analysis" counts athlete-seasons that also meet the eligibility rule.

| Event | Athlete-seasons (verified) | In analysis |
|---|---|---|
| 4 min | 996 | 667 |
| 30 min | 5,840 | 2,190 |
| 60 min | 2,737 | 1,294 |
| 500 m | 3,761 | 2,021 |
| 1000 m | 4,869 | 2,390 |
| 2000 m | 5,704 | 2,780 |
| 5000 m | 6,051 | 2,549 |
| 6000 m | 2,532 | 1,295 |
| 10,000 m | 5,610 | 2,260 |

**Eligibility.** An athlete-season is analysed if it has at least 3 events whose longest and shortest distances differ by a factor of at least 3, so that a slope is identifiable. Table 2 shows how the data narrow down.

Table 2: Data funnel.

| Step | Rows | Athlete-seasons |
|---|---|---|
| rows in CSV | 41,712 | 18,985 |
| rows with an athlete id | 38,435 | 18,985 |
| verified (individual) results | 38,103 | 18,903 |
| RowErg (static) only | 38,103 | 18,903 |
| valid time and distance | 38,103 | 18,903 |
| one result per athlete-season-event | 38,100 | 18,903 |
| >= 3 events and distance ratio >= 3 | 17,446 | 3,993 |

**Who is in the sample.** Only a minority of athlete-seasons qualify, and the sample is dominated by men and by masters rowers (Table 4). To see whether qualifying athletes are unusual, Table 3 compares the median 2 km time of qualifying athletes with all verified athletes who have a 2 km result.

Table 3: Median 2 km time of qualifying athletes against all verified athletes with a 2 km result.

| Group | Sex | n | Median 2 km (s) |
|---|---|---|---|
| all verified athletes with a 2 km result | F | 1,216 | 539.1 |
| athletes in the analysis (with a 2 km result) | F | 526 | 531.1 |
| all verified athletes with a 2 km result | M | 4,488 | 448.9 |
| athletes in the analysis (with a 2 km result) | M | 2,254 | 449.1 |

# 4 Methods

**Per-athlete exponents.** Each athlete-season gets its own least-squares slope. We report their median and mean, and the share below $\beta$.

**Pooled within-athlete exponent.** The pooled slope is $\sum S_{xy} \,/\, \sum S_{xx}$ over athlete-seasons, where $S$ are within-athlete sums of squares and cross-products. It removes each athlete's overall level and weights athletes by the spread of their distances. Confidence intervals come from a cluster bootstrap (2,000 resamples) that resamples athletes, so an athlete with several seasons is not counted as independent.

**Distance dependence.** We estimate the quadratic model above, pooled and by group. Separately, for every event we compute the exponent implied between an athlete's 2000 m result (the anchor) and that event, $\ln(t_e \,/\, t_0) \,/ \ln(d_e \,/\, d_0)$, together with the percentage by which the actual time differs from the 1.064 prediction. These two analyses do not need the same athletes to have three events.

**Event mix.** Because curvature makes the slope depend on which events an athlete rowed, we repeat the age and sex comparison within fixed distance windows (1–10 km and 2–21 km), using the same eligibility rule with a distance ratio of at least 2.

**Effort sensitivity.** Not every logged result is an all-out effort. We re-estimate with (i) a Huber-robust regression, (ii) dropping the 10% or 25% of athlete-seasons that fit worst after removing the common curvature, (iii) efforts up to 6 km only, and (iv) distance events only. These variants show how much the answer depends on unusual results; they cannot prove that all remaining results are maximal.

# 5 Results

## 5.1 Overall exponent

Table 4 gives the pooled exponent for all athletes and by sex and age group. The pooled exponent for all athletes is 1.067 [1.066, 1.068] the median athlete slope is 1.067 with a standard deviation of 0.034 across athletes, and 44% of athletes have a slope below 1.064. Figure 1 shows the distribution.

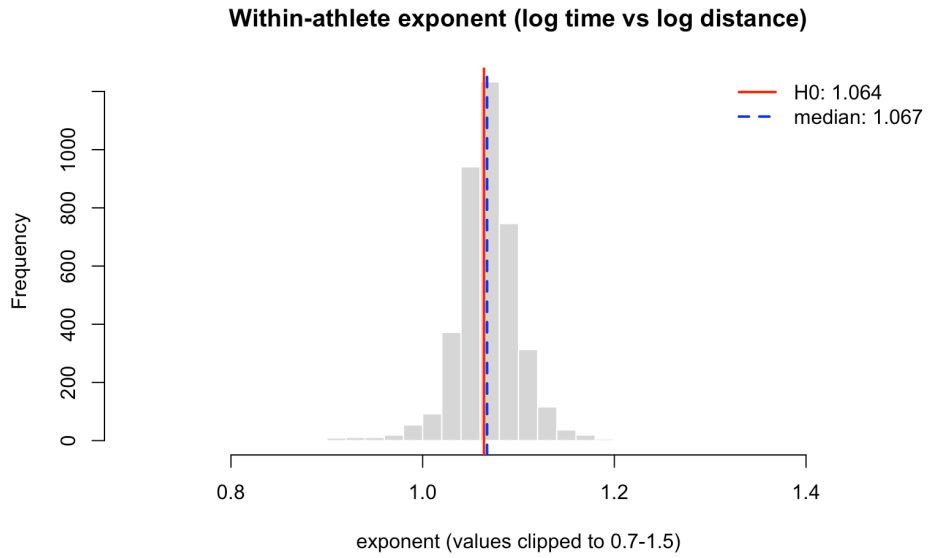



Figure 1: Within-athlete exponents. Red: hypothesised 1.064; blue: median.

## 5.2 Age and sex

Table 4: Within-athlete exponent by sex and age group. Groups with fewer than 30 athletes have no confidence interval.

| Group | Athletes | Median | Pooled | 95% CI | Exponent at 2 km |
|---|---|---|---|---|---|
| All athletes | 3,993 | 1.067 | 1.067 | [1.066, 1.068] | 1.071 |
| Women | 695 | 1.061 | 1.060 | [1.057, 1.063] | 1.061 |
| Men | 3,298 | 1.069 | 1.069 | [1.068, 1.070] | 1.073 |
| ≤18 | 99 | 1.083 | 1.082 | [1.076, 1.088] | 1.081 |
| 19–29 | 181 | 1.082 | 1.085 | [1.081, 1.090] | 1.088 |
| 30–39 | 638 | 1.076 | 1.077 | [1.075, 1.080] | 1.082 |
| 40–49 | 1,074 | 1.070 | 1.069 | [1.067, 1.071] | 1.073 |
| 50–59 | 1,039 | 1.065 | 1.064 | [1.062, 1.066] | 1.068 |
| 60+ | 962 | 1.060 | 1.060 | [1.058, 1.062] | 1.064 |
| Women ≤18 | 18 | 1.086 | – | [–, –] | – |
| Women 19–29 | 30 | 1.082 | 1.083 | [1.069, 1.097] | 1.081 |
| Women 30–39 | 95 | 1.071 | 1.071 | [1.066, 1.077] | 1.073 |
| Women 40–49 | 184 | 1.062 | 1.061 | [1.056, 1.066] | 1.062 |
| Women 50–59 | 163 | 1.059 | 1.056 | [1.049, 1.062] | 1.056 |
| Women 60+ | 205 | 1.055 | 1.054 | [1.050, 1.058] | 1.056 |
| Men ≤18 | 81 | 1.082 | 1.083 | [1.077, 1.089] | 1.082 |
| Men 19–29 | 151 | 1.083 | 1.086 | [1.081, 1.090] | 1.089 |
| Men 30–39 | 543 | 1.077 | 1.078 | [1.076, 1.081] | 1.083 |
| Men 40–49 | 890 | 1.071 | 1.070 | [1.068, 1.073] | 1.075 |
| Men 50–59 | 876 | 1.066 | 1.065 | [1.063, 1.067] | 1.070 |
| Men 60+ | 757 | 1.061 | 1.062 | [1.060, 1.064] | 1.067 |

The exponent falls with age and is lower for women than for men (Table 4, Figure 2). Because older athletes tend to row longer events and curvature makes slopes depend on the event mix, we checked whether any difference could be an artefact. Table 5 repeats the comparison inside fixed distance windows, and the last column of Table 4 gives the exponent at 2 km from the common-shape quadratic model.

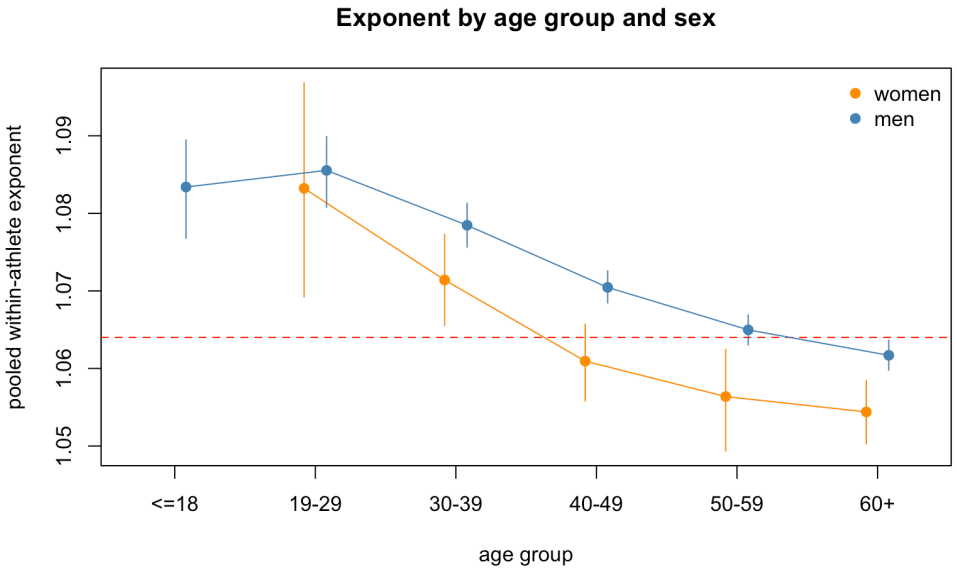



Figure 2: Pooled exponent by age group and sex (95% cluster-bootstrap intervals).

Table 5: Pooled exponent for the same athletes grouped by age, in fixed distance windows.

| Group | All events | 1–10 km window | 2–21 km window | Exponent at 2 km |
|---|---|---|---|---|
| All athletes | 1.067 | 1.060 | 1.063 | 1.071 |
| Women | 1.060 | 1.056 | 1.061 | 1.061 |
| Men | 1.069 | 1.061 | 1.064 | 1.073 |
| ≤18 | 1.082 | 1.081 | 1.093 | 1.081 |
| 19–29 | 1.085 | 1.079 | 1.084 | 1.088 |
| 30–39 | 1.077 | 1.068 | 1.071 | 1.082 |
| 40–49 | 1.069 | 1.061 | 1.065 | 1.073 |
| 50–59 | 1.064 | 1.057 | 1.059 | 1.068 |
| 60+ | 1.060 | 1.054 | 1.054 | 1.064 |

Inside both windows the age gradient persists: in the 2–21 km window the pooled exponent goes from 1.093 (under 19) to 1.054 (60 and over). Women minus men is −0.005 in the 1–10 km window and −0.002 in the 2–21 km window. So the age difference is not explained by event choice alone. We have not tested why it arises; training history, event specialisation and body mass (not available in the rankings) are all candidates.

## 5.3 Distance dependence

The quadratic model gives a local exponent of 1.071 at 2 km and a curvature of −0.0072 [−0.0081, −0.0064]. Using that curved model instead of a constant 1.064 changes predicted times relative to the 2 km time as in Table 6.

Table 6: Predicted time as a multiple of the 2 km time.

| Distance | Constant 1.064 | Curved model | Difference |
|---|---|---|---|
| 500 m | 0.229 | 0.223 | −2.3% |
| 1000 m | 0.478 | 0.474 | −0.8% |
| 5000 m | 2.651 | 2.652 | 0.0% |
| 6000 m | 3.219 | 3.215 | −0.1% |
| 10000 m | 5.542 | 5.500 | −0.8% |

Figure 3 and Table 7 show the exponent implied between the 2 km anchor and each other event. A value near 1.064 means that event is consistent with the hypothesis relative to the 2 km result.

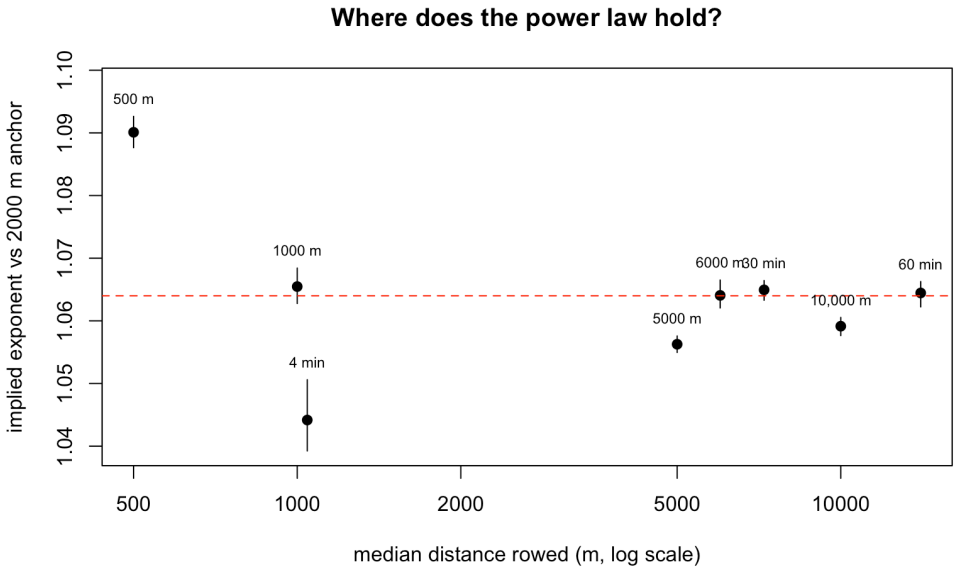



Figure 3: Exponent implied between the 2 km result and each other event (median with 95% bootstrap interval); red line: 1.064.

Table 7: Implied exponent relative to the 2 km result. Positive percentages mean slower than predicted.

| Event | n | Implied exponent | 95% CI | Median time vs 1.064 prediction |
|---|---|---|---|---|
| 500 m | 1,480 | 1.090 | [1.088, 1.093] | −3.6% |
| 1000 m | 1,757 | 1.065 | [1.063, 1.068] | −0.1% |
| 4 min | 430 | 1.044 | [1.039, 1.051] | 1.2% |
| 5000 m | 2,336 | 1.056 | [1.055, 1.058] | −0.7% |
| 6000 m | 925 | 1.064 | [1.062, 1.067] | 0.0% |
| 30 min | 1,615 | 1.065 | [1.063, 1.066] | 0.1% |
| 10,000 m | 1,697 | 1.059 | [1.058, 1.061] | −0.8% |
| 60 min | 859 | 1.064 | [1.062, 1.066] | 0.1% |

The implied exponent ranges from 1.044 (4 min) to 1.090 (500 m), so how well the hypothesis holds depends on which event is compared with the 2 km result.

## 5.4 Distance windows

Table 8: Pooled exponent in different distance windows.

| Window (distance rowed) | Athletes | Pooled exponent | 95% CI |
|---|---|---|---|
| all events | 4,555 | 1.067 | [1.066, 1.068] |
| 500 m – 10 km | 4,207 | 1.067 | [1.066, 1.068] |
| 1 km – 10 km | 3,468 | 1.060 | [1.059, 1.061] |
| 2 km – 21 km | 2,930 | 1.063 | [1.062, 1.065] |
| ≥ 2 km | 2,930 | 1.063 | [1.062, 1.065] |
| ≤ 6 km | 2,471 | 1.070 | [1.069, 1.072] |

The pooled exponent varies with the window by about 0.010 (Table 8), which is more than the gap between the overall estimate and 1.064. The hypothesis is therefore best stated for a defined range of distances rather than as a universal constant.

## 5.5 Effort sensitivity and other checks

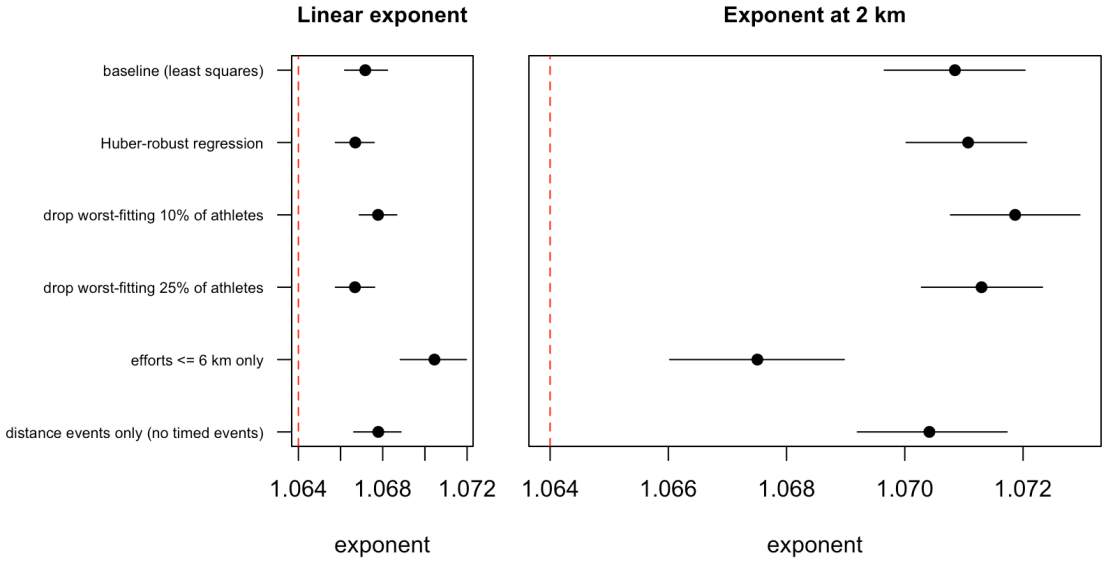



Figure 4: Linear exponent (left) and exponent at 2 km from the quadratic fit (right) under different treatments of unusual results. Red: 1.064.

Table 9: Sensitivity of the estimates to unusual or sub-maximal results.

| Variant | Athletes | Linear exponent | 95% CI | Exponent at 2 km | Curvature |
|---|---|---|---|---|---|
| baseline (least squares) | 3,993 | 1.067 | [1.066, 1.068] | 1.071 | −0.0072 |
| Huber-robust regression | 3,993 | 1.067 | [1.066, 1.068] | 1.071 | −0.0074 |
| drop worst-fitting 10% of athletes | 3,593 | 1.068 | [1.067, 1.069] | 1.072 | −0.0078 |
| drop worst-fitting 25% of athletes | 2,995 | 1.067 | [1.066, 1.068] | 1.071 | −0.0086 |
| efforts ≤ 6 km only | 2,471 | 1.070 | [1.069, 1.072] | 1.068 | −0.0142 |
| distance events only (no timed events) | 3,234 | 1.068 | [1.067, 1.069] | 1.070 | −0.0094 |

The baseline and the alternatives differ by at most 0.003 in the linear exponent (Table 9). The residual scatter of an athlete-season around its fitted curve has a median of 1.82% per result, a 90th percentile of 5.39% and a 99th percentile of 12.89%. The skewness of the pooled residuals is 1.41, which points to a tail of unusually slow results.

Requiring more events per athlete-season (Table 10) moves the pooled exponent by at most 0.002, which suggests the main result is not driven by how athletes qualify.

Table 10: Pooled exponent when more events per athlete-season are required.

| Minimum events | Athletes | Pooled exponent | 95% CI |
|---|---|---|---|
| ≥ 2 events | 5,303 | 1.068 | [1.067, 1.069] |
| ≥ 3 events | 3,993 | 1.067 | [1.066, 1.068] |
| ≥ 4 events | 2,465 | 1.066 | [1.065, 1.067] |
| ≥ 5 events | 1,440 | 1.066 | [1.065, 1.067] |

# 6 Discussion

**Sub-maximal efforts.** The hypothesis describes maximal efforts, but a logged verified result can be an easy piece. Because each ranking keeps an athlete's best verified result for an event and season, repeated easy sessions are mostly filtered out, but an athlete whose only verified 10 km is a steady row still contributes it. The sign of the resulting bias depends on where it occurs: sub-maximal results in long events raise the implied exponent; in short events they lower it; and a sub-maximal 2 km anchor shifts every implied exponent. These effects add noise (part of the athlete-to-athlete spread of slopes is probably effort rather than physiology) and could shift the pooled values in either direction. The sensitivity analyses suggest the headline estimates are not driven by obviously unusual results, but they cannot identify results that are consistently easy. The estimates should be read as exponents of typical verified efforts rather than of strictly maximal ones.

**Selection.** Athletes choose their events. Only about 21% of verified athlete-seasons have three or more events across a threefold range of distances, and the analysed sample is 83% male and 77% aged 40 or over. Qualifying athletes can be compared with all verified athletes by their 2 km time (Table 3), but they may still differ in motivation and training specificity. Results for women (695 athletes) and for athletes under 30 rest on much smaller groups.

**The 4-minute test.** The implied exponent for 4 minutes (1.044) is clearly lower than for the 1000 m event (1.065), although the two efforts last nearly the same time. This suggests the people who record verified 4-minute results are a different group, or that timed and distance results are not equivalent; it deserves a separate look before being interpreted physiologically.

**Range.** We excluded the extremes by design, so conclusions apply to roughly 500 m to 10 km (about 1.5 to 60 minutes), with the 60-minute event reaching about 14 km. The finding that the local exponent changes with distance means extrapolation to the marathon or to 100 m would not be safe.

**Other limitations.** One season is analysed. Body mass, drag factor and sleeping/training state are not available. Race-verified results were excluded because they cannot be linked to athletes. Age is as recorded in the ranking. Confidence intervals reflect sampling variation only, not model error.

# 7 Conclusion

For verified RowErg results from 2026 the within-athlete exponent is 1.067 on average ([1.066, 1.068]), close to the 1.064 used so far. Using 1.064 instead changes a predicted 6 km time by 0.3% relative to the 2 km time. But the exponent is not a universal constant: it falls with age, is lower for women and decreases with distance. A prediction rule that uses the 2 km exponent (1.071) with the curvature term, or one specific to age and sex, should do better than a single value; we did not test prediction accuracy directly. Whether efforts are all maximal remains the main uncontrolled factor.

# 8 Appendix: reproducibility

The report is produced by two scripts. `concept2_rankings.R` downloads and caches the ranking lists (with `status=verified` and `rower=rower`) into `data/rankings_long.csv`; `power_law_analysis.R` reads that file and writes `results/` (tables, `key_numbers.json`, figures). Settings: minimum 3 events, distance ratio at least 3, anchor event 2000 m, 2,000 bootstrap resamples, exponent under test 1.064. Compile this report with `typst compile report.typ`.

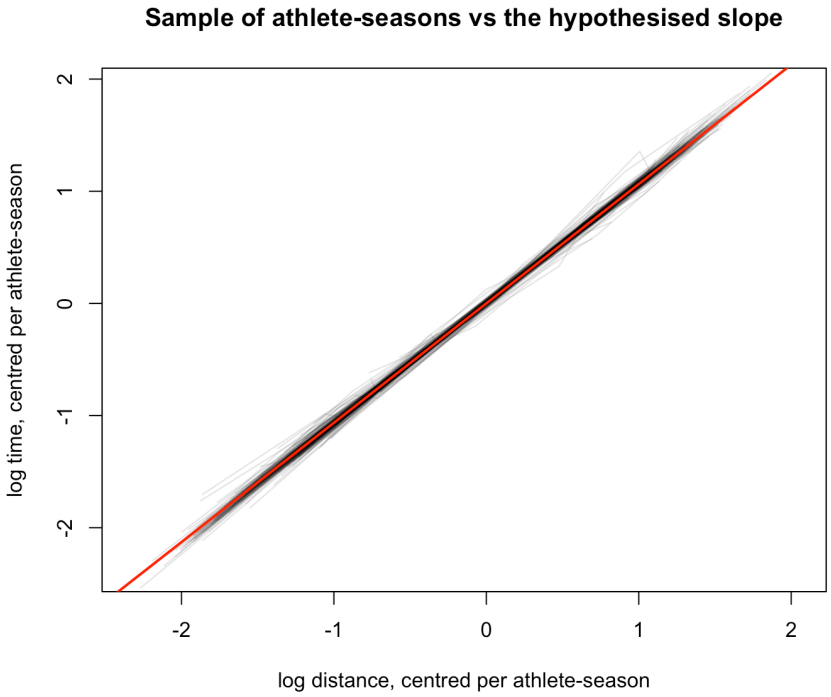



Figure 5: A random sample of athlete-seasons: log time against log distance, centred per athlete-season. The red line has the hypothesised slope.